How to read academic papers

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Let's get these points out of the way first: Paper reading, unlike reading a textbook, doesn't have to be linear. You don't have to understand all the details in a paper. It's a successful session if after reading, you can answer 3 things: **the *What, Why, and How***. What are they?

What

- What is the paper trying to solve? (aka the research questions)
- What are the key concepts, models, or methods?
- What are the main findings/results?

Why

- Why does this problem matter? Why should anyone care? (aka the significance)
- Why did the authors choose this method/dataset/theory?
- Why is this contribution important compared to prior work?
- Why should I trust the results?

How

- How did the author(s) answer the question? (what was the study design?)
- How does the evidence support the claims? (method/dataset/theory/experiments/analyses)
- How does this paper connect to existing literature?

Even though the papers in this course were curated for you with some intentions and purposes, each reader might have very different goals when reading a paper. Once you've identified the goals, they will guide how much emphasis you place on different parts of the paper. What has been working so far for me is as follows:

1) Decide the goals before reading. Why do I want to learn about this? What do I already know about this topic? What do I expect the authors' main claim to be?

2) Skimming: This gives a high-level view of the paper. Read the Abstract and Conclusion. During this stage, keep track of the things that are interesting, unexpected, simply new — while keeping in mind the above *What, Why, and How.*

3) Close read: Read the rest of the paper, trying to answer/revise your answers to the *What, Why, and How* questions.

4) Takeaways. Try to answer at least 4 out of these 6 questions in about 500-1000 words.

- What are the 3 most important ideas from this paper?
- Do I agree with the conclusions? Why/why not?
- What assumptions does the paper make?
- What are the strengths/ weaknesses of the work?
- What questions remain unanswered? What could be improved or investigated further?
- How does this paper influence my understanding of the field? What can I apply to my own research?

This was my anecdote but there are many guides for paper reading out there, for example: Ten simple rules for reading a scientific paper. Happy reading!